

ADVENTUREWORKS

EXECUTIVE SALES ANALYTICS SOLUTION

Power BI End-to-End Business Intelligence Case Study

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\$24.9M

Revenue

\$10.5M

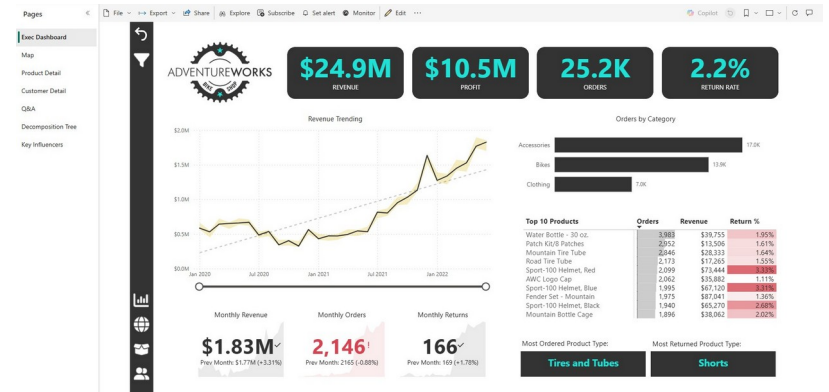
Profit

25.2K

Orders

2.2%

Return Rate



Executive Summary

- Portfolio Power BI solution using the AdventureWorks dataset.
- Demonstrates the full BI lifecycle: requirements, data preparation, modeling, DAX, design, validation, and insight delivery.
- Built to help leadership monitor sales, profit, orders, returns, regional performance, product trends, and customer behavior.
- Designed as a decision-support dashboard, not only a visual report.

THE SITUATION

The management team at **AdventureWorks** needs a reporting solution to monitor business performance and make data-driven decisions.

THE BRIEF

The team requires **insights** to:

- ✓ Track key **KPIs** (*sales, revenue, profit, returns*)
- ✓ Compare **regional performance**
- ✓ Analyze **product-level trends**
- ✓ Identify **high-value customers**



THE OBJECTIVE

Use **Power BI Desktop** to:

- ✓ Connect and transform the raw data
- ✓ Build a relational data model
- ✓ Create calculated columns and measures with DAX
- ✓ Design an interactive dashboard to visualize the data



Business Problem

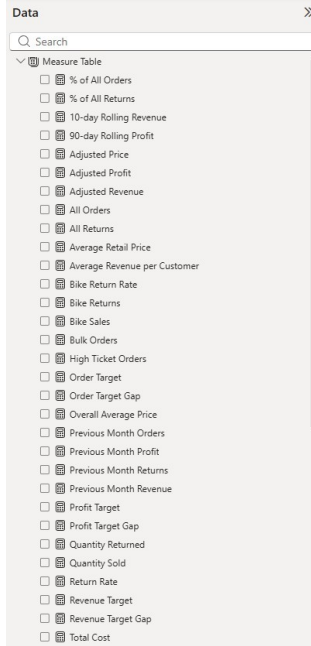
Key Business Questions	Data Sources Used
• How are revenue and profit trending over time? • Which product categories and products drive order volume? • Which regions and countries perform best? • Who are the highest-value customers? • How would product price changes affect profit? • What factors influence customer behavior and pricing?	• Sales transactions: revenue, orders, quantity, cost, profit. • Returns data: return quantity, return rate, product returns. • Customer lookup: customer profile, income, occupation, segmentation. • Product lookup, category, and subcategory tables. • Territory lookup: region and country analysis. • Calendar lookup: trends, previous month, rolling metrics, YTD.

Star Schema Model



- Used Sales Data as the main transaction fact table.
- Used Returns Data as a separate fact table for return behavior analysis.
- Connected Calendar Lookup for monthly, rolling, and YTD analysis.
- Connected Customer and Territory dimensions for segmentation and regional reporting.
- Connected Product, Category, and Subcategory tables to support product hierarchy reporting.

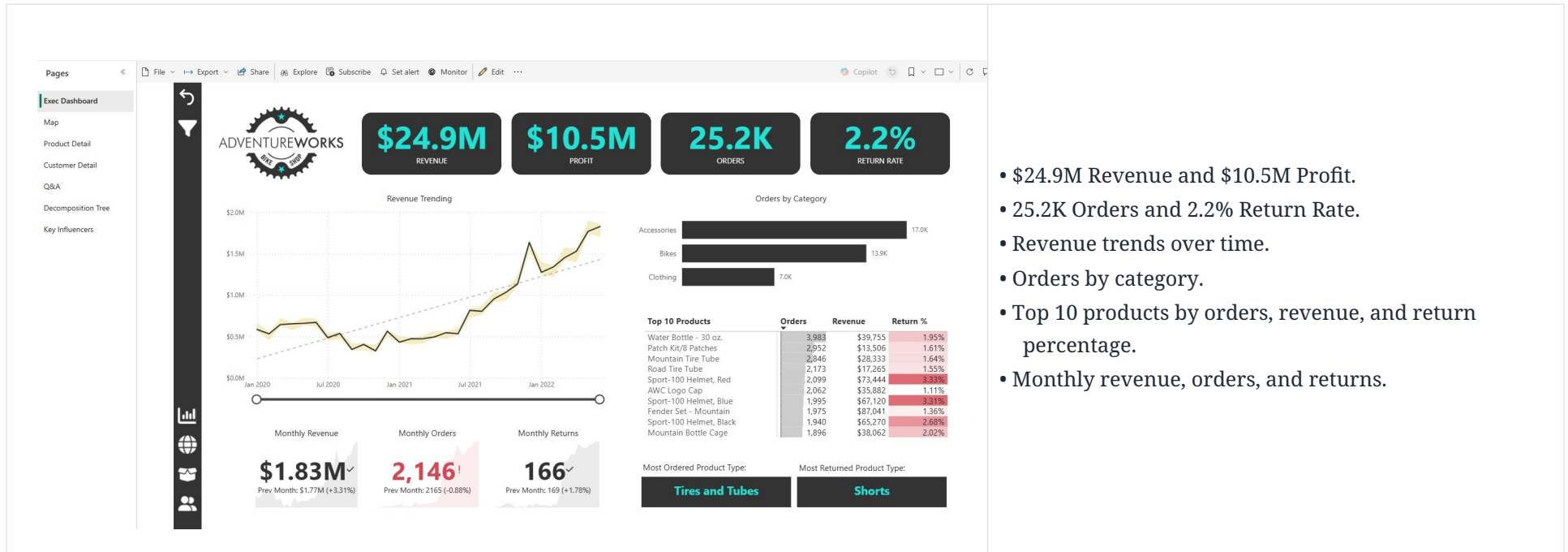
DAX Development



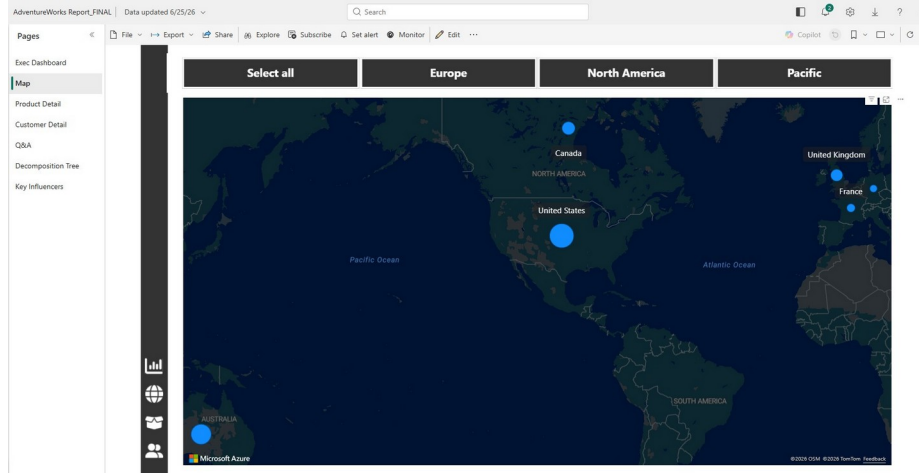
- Financial KPIs: revenue, profit, cost, average retail price.
- Operational KPIs: orders, returns, return rate, weekend orders.
- Time intelligence: YTD revenue, rolling revenue/profit, previous month metrics.
- Target analysis: revenue, profit, order targets and gaps.
- Customer metrics: total customers and average revenue per customer.
- Scenario analysis: adjusted price, adjusted revenue, adjusted profit.

- Converts transaction-level data into business-ready KPIs.
- Supports monthly performance monitoring.
- Allows actuals vs. target comparison.
- Enables pricing scenario analysis.
- Provides consistent calculations across report pages.

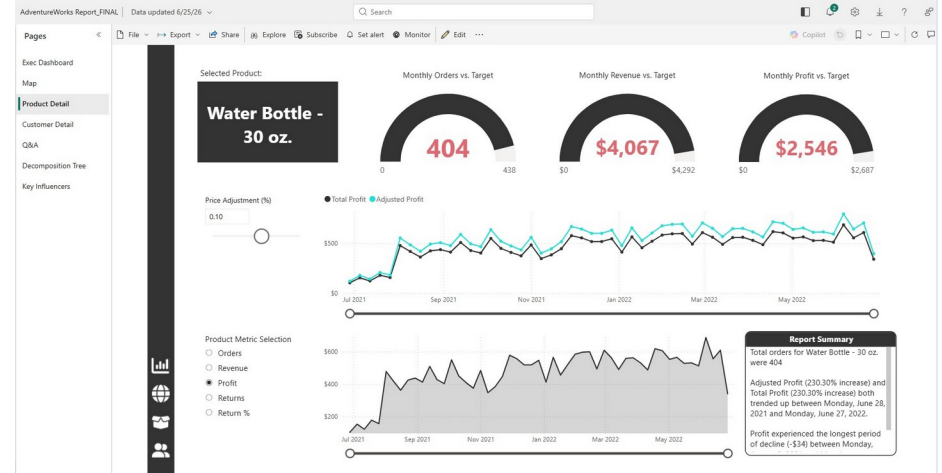
Executive Dashboard



Regional Performance and Product Detail



Regional Analysis: compares North America, Europe, and Pacific performance to support market prioritization.



Product Detail: analyzes product-level orders, revenue, profit, targets, and what-if price adjustment scenarios.

The image displays two screenshots of the AdventureWorks report interface, illustrating different analytical views of customer and product data.

Left Screenshot: Customer Analysis

This view provides a comprehensive overview of customer performance. The sidebar on the left lists navigation options: Exec Dashboard, Map, Product Detail, **Customer Detail**, Q&A, Decomposition Tree, and Key Influencers. The main content area features:

- KPI Cards:** Two prominent cards at the top left show "17.4K UNIQUE CUSTOMERS" and "\$1,431 REVENUE PER CUSTOMER".
- Line Chart:** A chart titled "Total Customers" and "Revenue per Customer" shows trends from January 2020 to January 2022.
- Donut Charts:** Two donut charts on the left represent "Orders by Income Level" (High 2.8K, Average 11.9K, Low 10.2K) and "Orders by Occupation" (Management 6.4K, Professional 7.9K, Skilled Manual 6.0K).
- Table:** A table titled "Top 100 Customers" lists customer details, including Customer Key, Full Name, Orders, and Revenue. The top customer is Mr. Maurice Shan with 6 orders and \$17,408 revenue.
- Summary Card:** A card for "Mr. Maurice Shan" shows he has 6 orders and generated \$12.4K in revenue.
- Insight:** A note at the bottom states: "Among customers in skilled manual roles in 2022, Ruben Suarez drove the most revenue at \$4,683."

Right Screenshot: Decomposition Tree

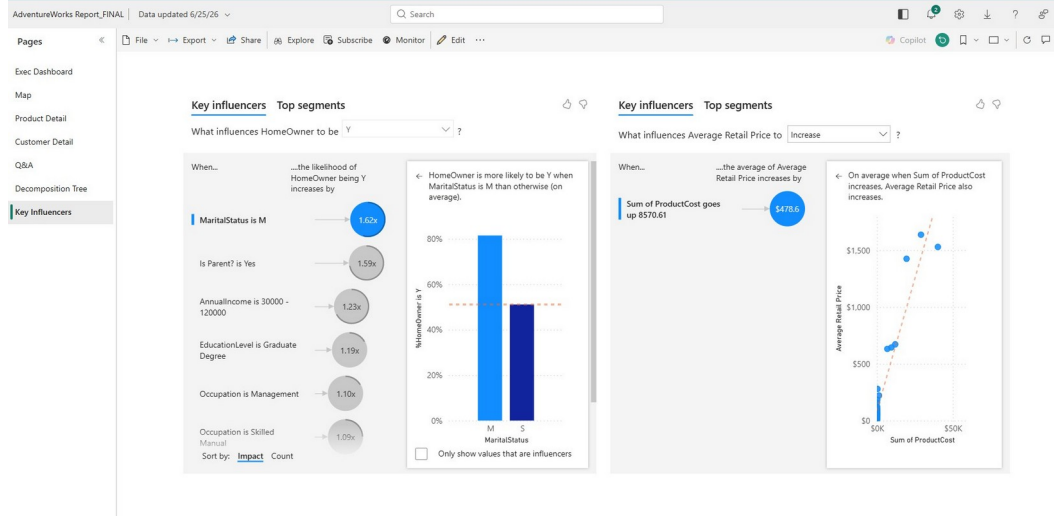
This view breaks down the total orders into a hierarchical structure. The sidebar on the left lists navigation options: Exec Dashboard, Map, Product Detail, Customer Detail, Q&A, **Decomposition Tree**, and Key Influencers. The main content area features:

- Tree Structure:** A tree diagram on the left shows the hierarchy: Total Orders (9,084) -> Accessories (16,983) -> Tires and Tubes (1,084) -> ProductName (1,084).
- Summary Card:** A large card on the right displays "9,084 Total Orders".
- Product Details:** A table on the right lists product details, including ProductName, SubcategoryName, and ProductKey. The top product is Patch K&U Patches with a product key of 2,912.

Customer Analysis: identifies high-value customers, revenue per customer, customer segments, income level, and occupation trends.

Decomposition Tree: breaks total orders into category, subcategory, and product-level drivers for root-cause style exploration.

Key Influencers and Quality Checks



- Checked revenue and profit KPI totals against model-level calculations.
- Validated order and return counts across executive and product pages.
- Tested relationship filtering between lookup tables and fact tables.
- Verified monthly trends, previous month metrics, rolling metrics, and YTD behavior.
- Tested slicers, navigation, drill behavior, and cross-filtering.
- Confirmed visual consistency when filters were applied.

Business Impact

- Tracks revenue, profit, orders, returns, and return rate in one executive dashboard.
- Identifies top products, category performance, return issues, and product trends.
- Compares regional performance and highlights geographic differences.
- Identifies high-value customers and customer segment patterns.
- Supports what-if pricing analysis and AI-assisted insight discovery.

- Imported and prepared raw CSV files in Power BI.
- Built Power Query transformations and lookup tables.
- Designed the Star Schema model and relationships.
- Created 40+ DAX measures for KPIs, targets, trends, and scenarios.
- Built executive, regional, product, customer, decomposition tree, and key influencer pages.
- Validated KPIs, filters, relationships, and cross-page consistency.

Skills Demonstrated

- | | |
|--|--|
| <ul style="list-style-type: none">• Power BI Desktop and Power BI Service.• Power Query data preparation and transformation.• Star Schema data modeling and relationship design.• DAX measures for financial KPIs, time intelligence, target analysis, and what-if analysis.• Dashboard design, interactivity, filters, slicers, drilldown, and AI visuals.• Data validation and business-focused storytelling. | <ul style="list-style-type: none">• Microsoft Certified: Power BI Data Analyst Associate (PL-300).• Experience in SQL, data validation, reporting, analytics support, and stakeholder communication.• Background includes enterprise healthcare data support, root cause analysis, dashboarding, and data-driven decision support.• LinkedIn: www.linkedin.com/in/belal-abu-khadejah-08a4b0198• GitHub: github.com/BelalKadejah |
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